

Simulated Ebola Outbreak

Overview

This is a simulated Ebola Virus Disease outbreak dataset, designed with properties mimicking the West African 2014–2015 epidemic. Created by Pierre Nouvellet as a teaching resource, it contains intentional data quality issues (e.g., negative incubation periods from erroneous date entries) to illustrate real-world data cleaning challenges. A cleaned version (`ebola_sim_clean`) with erroneous dates removed is also provided.

The dataset is a `NamedTuple` with two components: a linelist of cases and a contacts tracing `DataFrame`.

Loading the data

```
using DataFrames
using Outbreaks
data = Outbreaks.ebola_sim()
linelist = data.linelist
contacts = data.contacts
println("Linelist: $(size(linelist))")
println("Contacts: $(size(contacts))")
```

Linelist: (5888, 11)

Contacts: (3800, 3)

Linelist — first few rows

```
first(linelist, 6)
```

	case_id	generation	date_of_infection	date_of_onset	date_of_hospitalisation	
	String7	Int64	String15	Date	Date	
1	d1fafd	0	NA	2014-04-07	2014-04-17	...
2	53371b	1	2014-04-09	2014-04-15	2014-04-20	...
3	f5c3d8	1	2014-04-18	2014-04-21	2014-04-25	...
4	6c286a	2	NA	2014-04-27	2014-04-27	...
5	0f58c4	2	2014-04-22	2014-04-26	2014-04-29	...
6	49731d	0	2014-03-19	2014-04-25	2014-05-02	...

Linelist summary

```
describe(linelist)
```

	variable	mean	min	median	max	nmissing	
	Symbol	Union...	Any	Any	Any	Int64	
1	case_id		00031d		fff9d	0	...
2	generation	16.5617	0	16.0	37	0	...
3	date_of_infection		2014-03-19		NA	0	...
4	date_of_onset		2014-04-07	2014-10-21	2015-04-30	0	...
5	date_of_hospitalisation		2014-04-17	2014-10-23	2015-04-30	0	...
6	date_of_outcome		2014-04-19		NA	0	...
7	outcome		Death		Recover	0	...
8	gender		f		m	0	...
9	hospital		Connaught Hospital		other	0	...
10	lon	-13.2338	-13.2728	-13.2291	-13.2052	0	...
11	lat	8.46964	8.44621	8.46899	8.49175	0	...

Contacts — first few rows

```
first(contacts, 6)
```

	infector	case_id	source
	String7	String7	String7
1	d1fafd	53371b	other
2	cac51e	f5c3d8	funeral
3	f5c3d8	0f58c4	other
4	0f58c4	881bd4	other
5	8508df	40ae5f	other
6	127d83	f547d6	funeral

Cleaned version

```
clean = Outbreaks.ebola_sim_clean()
clean_ll = clean.linelist
println("Raw linelist rows: $(nrow(clean_ll))")
println("Cleaned linelist rows: $(nrow(clean_ll))")
println("Rows removed: $(nrow(clean_ll) - nrow(clean_ll))")
```

```
Raw linelist rows: 5888
Cleaned linelist rows: 5829
Rows removed: 59
```

Epidemic curve

```
using Dates
onset = dropmissing(clean_ll, :date_of_onset)
sort!(onset, :date_of_onset)
weekly = combine(groupby(transform(onset, :date_of_onset => (x -> x .- Dates.Day.(Dates.dayofweek.(x)
sort!(weekly, :week_start)
for row in eachrow(weekly)
    bar = repeat("█", max(1, round{Int, row.cases / 2}))
    println("$(row.week_start) | $bar $(row.cases)")
end
```

```
2014-04-07 | █ 1
2014-04-14 | █ 1
2014-04-21 | █ 5
2014-04-28 | █ 4
2014-05-05 | █ 12
2014-05-12 | █ 17
2014-05-19 | █ 15
```

